THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 2

Chapter 8: **Heredity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Chromosomes that determine sex are called 1

(a) genes **(b)** sex chromosomes **(c)** autosomes **(d)** plasmids

2. The tall x dwarf F1 plants are 1

(a) homozygous dwarf **(b)** heterozygous tall

(c) homozygous tall **(d)** pure dwarf

3. Chromosomes are made of 1

(a) cellulose **(b)** DNA and protein **(c)** only lipids **(d)** only starch

4. The egg of a human female contains 1

(a) no sex chromosomes **(b)** two X chromosomes

(c) one X chromosome as a sex chromosome **(d)** one Y chromosome

5. The F2 ratio for a monohybrid cross is 1

(a) 9 : 3 : 3 : 1 **(b)** 2 : 1 **(c)** 1 : 3 **(d)** 3 : 1

6. Variations among individuals are 1

(a) absent in humans **(b)** differences in characters

(c) not inherited **(d)** always harmful

7. A sperm can contain 1

(a) only Y **(b)** only X **(c)** XX **(d)** X or Y

8. The F2 ratio of the dihybrid cross is 1

(a) 1 : 2 : 1 **(b)** 1 : 1 **(c)** 9 : 3 : 3 : 1 **(d)** 3 : 1

9. In pea plants, the dominant trait for stem height is 1

(a) tall **(b)** medium

(c) intermediate height **(d)** dwarf

10. A child who is XX is 1

(a) a girl **(b)** a twin **(c)** a boy **(d)** neither

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is a gene? 2

12. What is an allele? 2

13. What is variation? 2

14. Name the sex chromosomes in a male and a female. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. What are autosomes? How many are present in a human cell? 3

16. Give three examples of traits that are inherited in humans. 3

17. State two ways to differentiate a trait that is inherited from one that is acquired. 3

OR

Distinguish between genotype and phenotype with an example. 3

18. Explain why the sex of a child cannot depend on the mother. 3

OR

What type of sperm gives a boy and what type gives a girl? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. How is sex determined in humans? 5

OR

Show by a Punnett square that a cross between two heterozygous tall plants gives dwarf plants. 5

20. What are inherited and acquired traits? Give two examples of each. 5

OR

Explain why the ratios 3 : 1 and 9 : 3 : 3 : 1 are significant. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND